

Smart_Food_Packaging & Cold_Chain_Risk_Analytics

High Risk Packages

2987

Compliance Rate

11

Total Packages

15,000

Cold Chain Failures

30%

Avg Risk Score

60%

Spoilage Rate

72%

Food Category

- ☐ Bakery
- ☐ Beverage
- ☐ Canned Food
- ☐ Dairy

Environmental Monitoring

Packaging & Quality...

Operational Analysis

Food Category by Avg Risk Score

Seafood	0.61
Frozen Food	0.61
Bakery	0.60
Beverage	0.60
Fresh Produce	0.60
Dairy	0.59
Meat Product	0.59

Storage Zone by Avg Risk Score

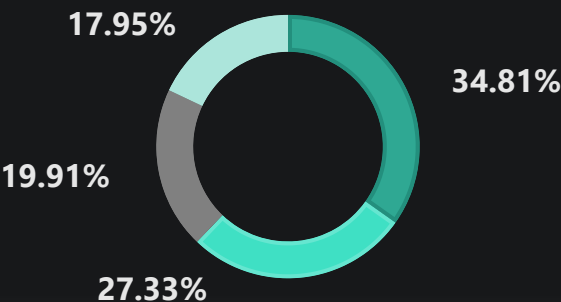


Storage Zone

- ☐ Cold Chain Storage
- ☐ Distribution Center
- ☐ Retail Shelf
- ☐ Transport Vehicle

Food Safety Risk Level Distribution

● Critical Risk ● Medium Risk ● High Risk ● Low Risk



Packaging Material by Avg Risk Score

Paperboard	0.61
PET Plastic	0.60
Glass Container	0.59
Vacuum Pouch	0.59
Composite Film	0.59
Aluminum Laminate	0.59
Biodegradable Film	0.59

Packaging Material

- ☐ Aluminum Laminate
- ☐ Biodegradable Film
- ☐ Composite Film

Food Safety Risk Level

- ☐ Critical Risk
- ☐ High Risk
- ☐ Low Risk
- ☐ Medium Risk

Smart_Food_Packaging & Cold_Chain_Risk_Analytics

Avg Shelf Life Left

9

Avg Freshness

57

Avg Temperature

7

Avg Humidity (%)

62

Temperature
Status

- ☐ High
- ☐ Low
- ☐ Medium
- ☐ Very High

Humidity
Status

- ☐ High
- ☐ Low
- ☐ Normal

Storage_Zone

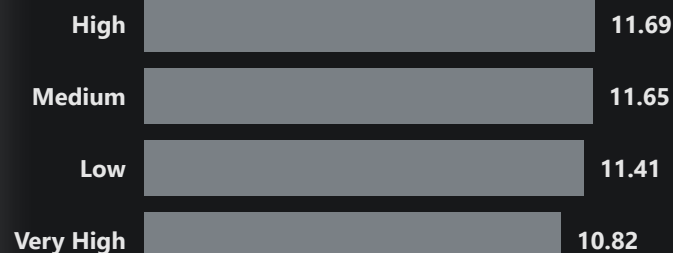
- ☐ Cold Chain S...
- ☐ Distribution ...
- ☐ Retail Shelf
- ☐ Transport Ve...
- ☐ Warehouse

Executive Overview

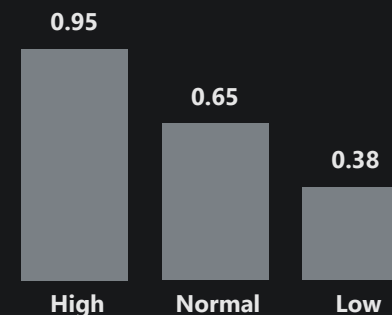
Packaging & Quality...

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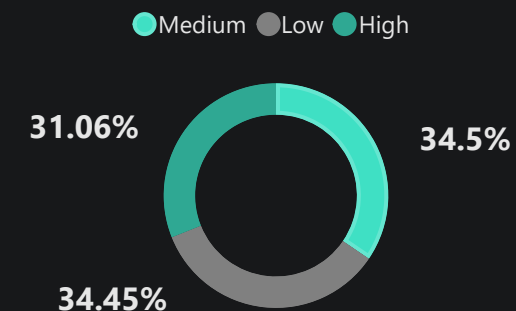
Temperature vs Shelf Life



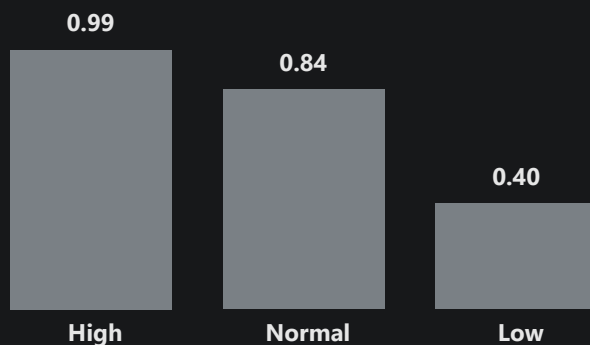
Humidity vs Risk Score



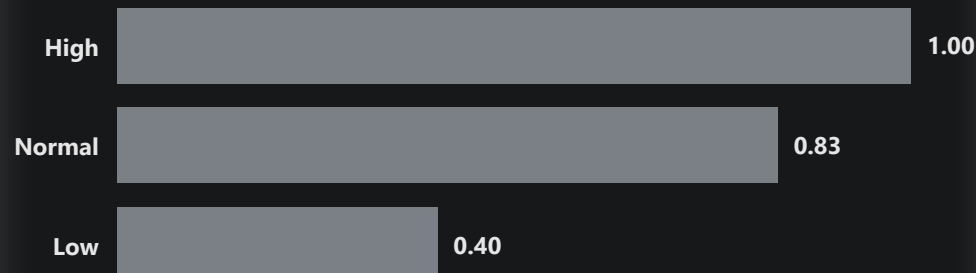
Light Exposure vs Freshness



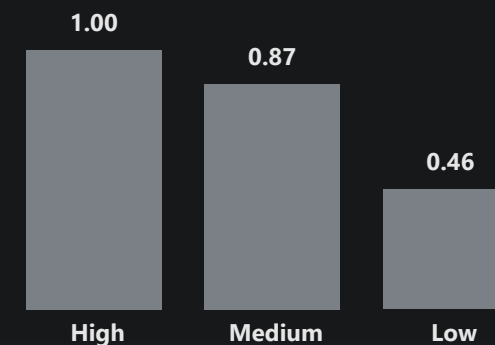
Oxygen Concentration vs Risk



Carbon Dioxide vs Risk



Volatile Gas vs Risk



Smart_Food_Packaging & Cold_Chain_Risk_Analytics

Avg Sensor Health

87

Damage Rate

55%

Packaging_Material

- ☐ Aluminum Laminate
- ☐ Biodegradable Film
- ☐ Composite Film
- ☐ Glass Container
- ☐ Paperboard
- ☐ PET Plastic
- ☐ Vacuum Pouch

Surface_Integrity_Status

- ☐ Contamination Defect
- ☐ Flawless Package
- ☐ Seal Defect

Digital_Twin_State

- ☐ Alert Twin State
- ☐ Critical Twin State

0.90

0.71

0.34

0.27

Criti...
Twin...

Alert
Twin...

War...
Twin...

Stable
Twin...

Compliance Status by Storage Zone

Compliance_Status ● Safe ● Unsafe

Transport Vehicle

10.57%

89.43%

Warehouse

11.34%

88.66%

Retail Shelf

10.69%

89.31%

Distribution Center

10.87%

89.13%

Cold Chain Storage

10.64%

89.36%

0%

50%

100%

Packaging Material vs Food Safety Risk

Paperboard

61.00%

PET Plastic

60.00%

Aluminum ...

59.00%

Biodegrad...

59.00%

Composite...

59.00%

Glass Cont...

59.00%

Vacuum Po...

59.00%

High Risk Packages by Storage_Zone

Cold Cha...

618

Distribut...

618

Retail Shelf

601

Transpor...

586

Warehouse

564

Packaging_Material

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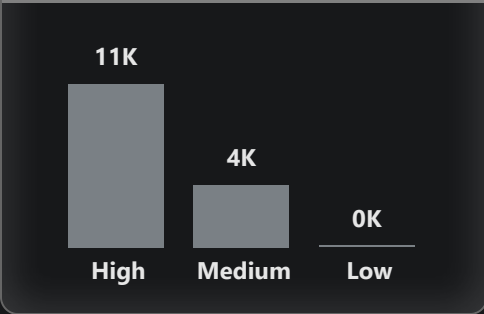
Smart_Food_Packaging & Cold_Chain_Risk_Analytics

Executive Overview

Environmental Monitoring

Packaging & Quality...

Transport Condition vs Spoilage Probability



Storage_Zone	Critical Risk	High Risk	Low Risk	Medium Risk	Total
Cold Chain Storage	1029	618	516	797	2960
Distribution Center	1006	618	535	823	2982
Retail Shelf	1042	601	522	829	2994
Transport Vehicle	1089	586	530	851	3056
Warehouse	1055	564	589	800	3008
Total	5221	2987	2692	4100	15000

Transport_Condition

☐ Frequent Vibration

☐ High Moisture Exposure

☐ Mild Shock

☐ Stable

☐ Temperature Fluctuation

Package_Size

☐ Bulk

☐ Large

☐ Medium

☐ Small

Storage_Zone

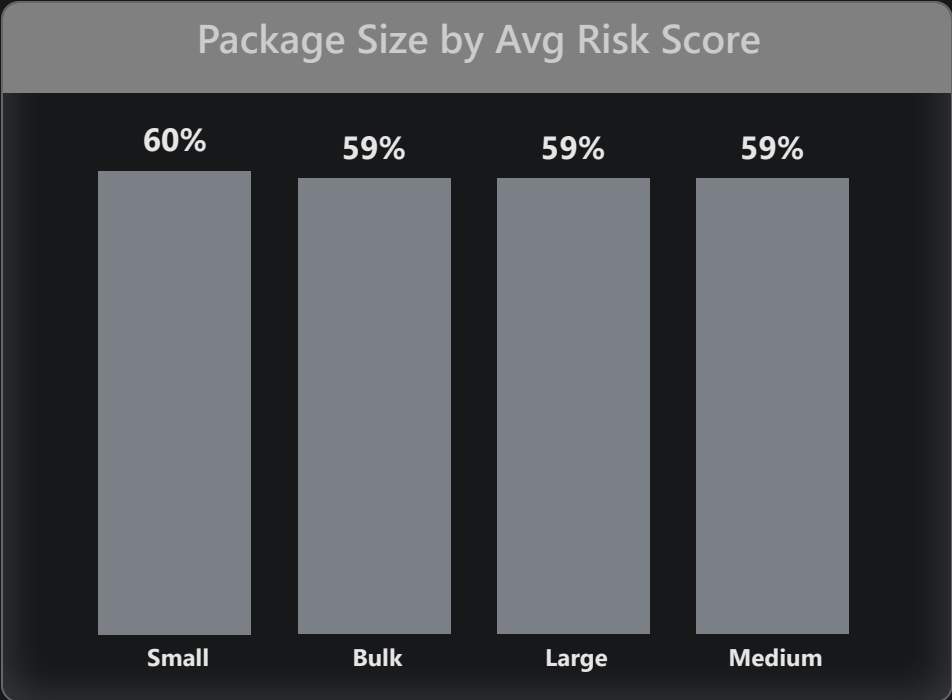
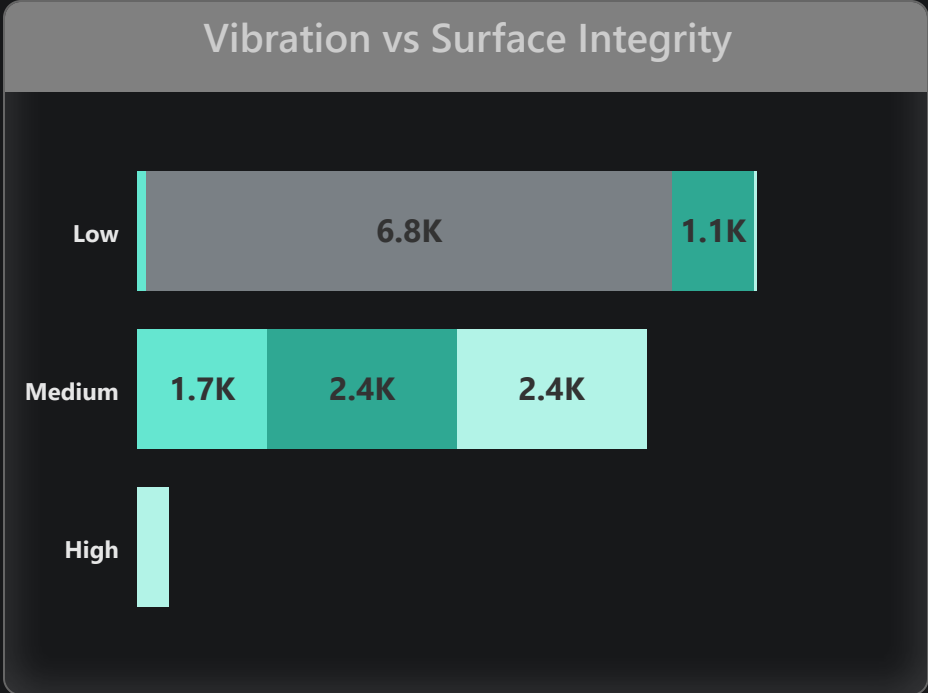
☐ Cold Chain Storage

☐ Distribution Center

☐ Retail Shelf

☐ Transport Vehicle

☐ Warehouse





Sample_ID	Food_Category	Packaging_Material	Storage_Zone	Package_Size	Food_Safety_Risk_Level	Compliance_Status	Digital_Twin_State	Sum of Temperature_C	Sum of Humidity_Percent	Sum of Oxygen_Percent
IPKG_01466	Frozen Food	Vacuum Pouch	Warehouse	Small	Critical Risk	Unsafe	Critical Twin State	16.87	95.00	
IPKG_01574	Ready Meal	Glass Container	Cold Chain Storage	Large	Critical Risk	Unsafe	Critical Twin State	7.60	95.00	
IPKG_02496	Meat Product	Biodegradable Film	Transport Vehicle	Bulk	Critical Risk	Unsafe	Critical Twin State	10.92	95.00	
IPKG_04084	Beverage	Paperboard	Retail Shelf	Large	Critical Risk	Unsafe	Critical Twin State	13.32	95.00	
IPKG_04393	Dairy	Glass Container	Retail Shelf	Bulk	Critical Risk	Unsafe	Critical Twin State	15.16	95.00	
IPKG_04563	Seafood	Glass Container	Warehouse	Medium	Critical Risk	Unsafe	Critical Twin State	13.89	95.00	
IPKG_06407	Fresh Produce	Biodegradable Film	Warehouse	Bulk	Critical Risk	Unsafe	Critical Twin State	14.64	95.00	
IPKG_06478	Meat Product	Composite Film	Transport Vehicle	Bulk	Critical Risk	Unsafe	Critical Twin State	11.53	95.00	
IPKG_06703	Frozen Food	Glass Container	Retail Shelf	Large	Critical Risk	Unsafe	Critical Twin State	14.46	95.00	
IPKG_07744	Canned Food	Paperboard	Transport Vehicle	Medium	Critical Risk	Unsafe	Critical Twin State	9.95	95.00	
IPKG_08501	Beverage	Biodegradable Film	Transport Vehicle	Medium	Critical Risk	Unsafe	Critical Twin State	10.38	95.00	
IPKG_10027	Seafood	Composite Film	Warehouse	Bulk	Critical Risk	Unsafe	Critical Twin State	9.24	95.00	
IPKG_10378	Meat Product	Glass Container	Distribution Center	Bulk	Critical Risk	Unsafe	Critical Twin State	17.04	95.00	
IPKG_11580	Meat Product	Paperboard	Distribution Center	Medium	Critical Risk	Unsafe	Critical Twin State	11.87	95.00	
IPKG_13089	Bakery	Biodegradable Film	Distribution Center	Bulk	Critical Risk	Unsafe	Critical Twin State	12.21	95.00	
IPKG_04696	Dairy	Aluminum Laminate	Transport Vehicle	Large	Critical Risk	Unsafe	Critical Twin State	14.83	94.99	
IPKG_12122	Dairy	Composite Film	Retail Shelf	Bulk	Critical Risk	Unsafe	Critical Twin State	8.82	94.89	
IPKG_01991	Dairy	Aluminum Laminate	Transport Vehicle	Medium	Critical Risk	Unsafe	Critical Twin State	6.71	94.70	
IPKG_08052	Seafood	Paperboard	Cold Chain Storage	Large	Critical Risk	Unsafe	Critical Twin State	12.18	94.70	
IPKG_01284	Meat Product	Biodegradable Film	Warehouse	Medium	Critical Risk	Unsafe	Critical Twin State	13.02	94.05	
IPKG_09632	Beverage	Aluminum Laminate	Cold Chain Storage	Medium	Critical Risk	Unsafe	Critical Twin State	10.97	94.03	
Total								102,257.42	934,059.21	